Graduate Research Plan Statement

Background: Most vertebrates have a retina with a dual photoreceptor system consisting of rods and cones and known as "duplex".[1] These photoreceptors undergo a photochemical reaction and convert the absorption of photons into electrical signals, which is then sent to the brain for further visual processing.[2] Rods are primarily used for dim light and monochromatic vision, while cones are used for colored vision and bright light conditions. In contrast to duplex retinas, the cartilaginous fish *Leucoraja erinacea* (Little Skate) is unusual in having a "simplex" retina with only a single type of rod-like photoreceptor.[3] In addition, previous studies have shown that photoreceptors in this retina can operate in both the rod and cone functional domains and cover almost the full range of environmental illumination.[3] We know little about the molecular and structural characteristics that mediates this plasticity in function. On the structural side, recent data from our group has shown that skate photoreceptors have a hybrid rod-cone structure.[4] For example, one hybrid feature that has been identified is the varying populations of synaptic ribbons across different skate photoreceptors. Ribbons are organelles exclusively found in sensory neurons that tether and organize synaptic vesicles in preparation for release of their neurotransmitter content into the synaptic cleft and onto post-synaptic neurons. The number of ribbons in vertebrate photoreceptors is stereotypical and is usually correlated with the number of post-synaptic processes that connect to a photoreceptor. However, it is still unknown if there is any correlation between the varying numbers of ribbons in skate photoreceptors, and the quantity of their post-synaptic contacts. On the molecular side, while rhodopsin is the only visual pigment identified to date in skate, we still do not know if there are any additional opsin genes, like cone-opsins, which might be expressed in skate retina and help mediate cone-like function. Our preliminary data suggests the existence of long-wavelength sensitive (LWS) opsin in the global genome of skate. Therefore, our goals are to: ***1) investigate the connectivity at the skate photoreceptor ribbon synapse, and 2) determine what molecular players mediate dual functionality in skate photoreceptors.*** *I hypothesize that 1) the number of ribbons increases with the number of post synaptic invaginations to sustain the frequency of neurotransmission, and 2) there is an additional cone-opsin, which mediates dual functionality.*

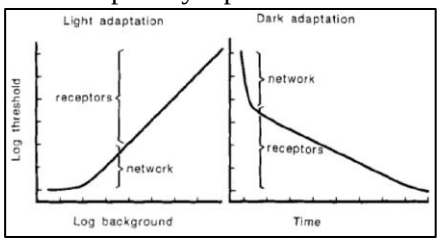


Figure 1: Little Skate Light and Dark Visual Adaptation Curve to Rhodopsin Bleaching[3]

Aim 1: ***To investigate the connectivity at the skate photoreceptor ribbon synapse.***

Rationale: Do the number of photoreceptor post-synaptic neurons require a specific number of photoreceptor ribbons to sustain neurotransmission in the skate simplex retina? The Anastassov lab generated preliminary data from 3D electron microscopy (EM) that described a hybrid morphology for ribbon synapses (***Fig. 2a-b***) in photoreceptors. Typically, in mammalian duplex retinas, the rod terminal has only one ribbon, while the cone terminal can have multiple ribbons (10-50).[5] Interestingly, in the skate photoreceptors, we observe a range of 1-4 ribbons distributed across terminals.[4] We are unsure why these photoreceptors have varying amounts of ribbons per terminal. Therefore, to observe the relationship between the number of ribbons and the number of post-synaptic neurons, I will quantify the number of postsynaptic invaginations photoreceptor terminals with different numbers of ribbons.

Experimental Design: ***Fig. 2a-c*** illustrates ribbon reconstructions and a brief outline of the experimental design. Serial block-face scanning electron microscopy (SBF-3DEM) will be performed on epoxy resin-embedded skate retinal tissue at the Cleveland Clinic 3DEM Ultrastructural Imaging and Computation Core. Reconstructions will be obtained from segmentations in Reconstruct and Amira software.

Anticipated Results: Preliminary data suggests an unusually high number of postsynaptic invaginations in skate photoreceptor terminals (>3-fold change compared to duplex retina rods).[4] I will use quantifications of ribbons and postsynaptic processes reconstructions to determine any correlation between these two structural features. Alternatively, if there is no correlation, then my data will indicate that the number of postsynaptic invaginations is independent from the number of ribbons.

Aim 2: ***To determine what molecular players mediate dual functionality in skate photoreceptors.***

Rationale: Previously observed hybrid rod-cone physiological features[4] in skate photoreceptors suggest that these photoreceptors might express cone-specific markers. Skate photoreceptors can function like cones or rods[3], but we do not know if they express any cone-specific visual pigments, or other closely related cone genes. Therefore, we will attempt to identify any cone-specific transcripts in our newly constructed skate retina cDNA library using a de novo assembly of a skate retinal transcriptome.

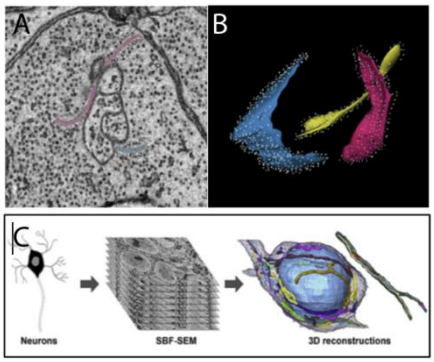


Figure 2: A) EM serial section of photoreceptor terminal. B) Reconstruction of individual ribbons (Blue, Pink, Yellow) from EM section. C) Outline of flowchart for reconstruction[6]

Experimental Design: We will look for cone-specific genes, like LWS opsin, in our retinal transcriptome using alignment to homologous sequences of closely related species. Our search will include other opsins, like short- and medium-wavelength sensitive opsins.[7] Potential candidates will be amplified from our cDNA library and used to develop probes for fluorescence *in situ* hybridization to detect mRNA transcripts for cone-specific transcripts. Alternatively, we can perform immunohistochemistry with cone-specific antibodies to detect spatial distribution of any cone proteins.

Anticipated Results: We anticipate detecting some type of cone-specific opsin isoform, because of the known dual functionality in skate photoreceptors and because of the presence of LWS in the global genome of skate. If we do not detect any cone-specific mRNA transcripts, this may indicate that daylight vision is not mediated through cone-opsin, or cone-specific genes. Therefore, we can explore alternative genes of interest related to rod functionality, like differentially tuned isoforms of rhodopsin.[8]

Feasibility: We already have preliminary SBF-3DEM data from the Cleveland Clinic and new fixed tissue has been sent out for serial sectioning. In addition, we have created a cDNA library from *L. erinacea* retina and performed chromogenic *in situ* hybridizations for several rod and bipolar cell mRNA transcripts. A standard protocol for *in situ* hybridization has been optimized, but we have not yet probed for any cone opsin transcripts and other cone specific markers.

Intellectual Merit: Exploring the structural and molecular hallmarks of a simplex retina can broaden our current knowledge of the vertebrate visual system. From our current analysis, we observed that ribbon synapse formation goes beyond the triad/tetrad formation in duplex retinas. Importantly, the formation of ribbon synapses in the simplex skate retina can help us understand the tradeoff between having dual photoreceptor systems and fewer postsynaptic partners, and single photoreceptor systems and more postsynaptic partners. In addition, our research will potentially identify genes that regulate the adaptation for daylight vision in photoreceptors from a simplex retina. In all, the *L. erinacea* model organism can offer new directions in understanding retinal circuitry.

Broader Impacts: This research can lead to potential translational studies exploring simplex visual systems for partial vision restoration during cone or rod degeneration. In addition, I plan to work with the engineering department at San Francisco State University to convert 3DEM data into tangible 3D models kits through 3D printing, which can be used for tactile learning in classroom settings. Akin to organic chemistry 3D model kits that help students understand structures, 3D models kits of cellular anatomy can make science accessible for blind and visually impaired students. These physical model kits can be new learning tools and help science instructors educate the blind and visually impaired. Tactile learning has the potential to offer new insights and help the understanding science from a perspective much different than that of previous generations in classroom settings. This type of research has inspired me to be involved in reaching out to the visually impaired community to introduce scientific research involving 3D reconstruction and the potential directions for the near future.

References: [1] Ebrey et al. *Prog Retin Eye Res.* 2001. 20(1):49-94. [2] Stryer. *Sci Amer*. 1987. 257(1):42-51. [3] Dowling et al. *J Exp. Zoology* 1990. 5:55-65. [4] Magaña-Hernández et al. *eNeuro. In Press.* 2023. [5] Mercer et al. *Vis Neurosci*. 2011. 28(6):453-471. [6] Faitg et al. *Cell Rep*. 2021. 109509. [7] Nathans. *Sci Amer.* 1989. 260(2): 42-49. [8] Musilova et al. *Science.* 2018. 364(6440): 588-592.